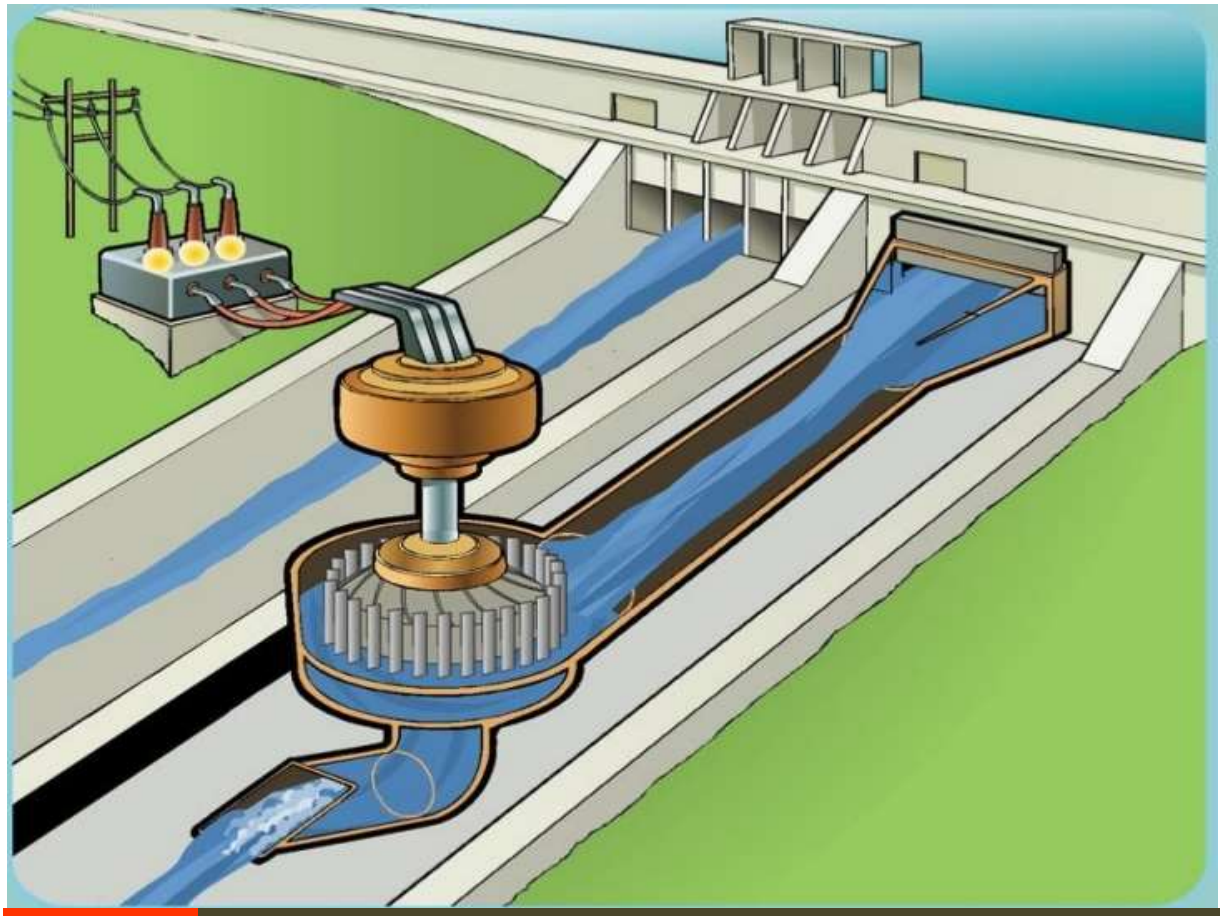


7 Alternating-Current Circuits and Electromagnetic Waves

Every time we turn on a television set, a stereo system, or any other electric appliances, we call on alternating currents (AC) to provide the power to operate them. We begin our study of AC circuits by examining the characteristics of a circuit containing a source of emf and one other circuit element: a resistor, a capacitor, or an inductor. Then we examine what happens when these elements are connected in combination with each other. Our discussion is limited to simple series configurations of the three kinds of elements.

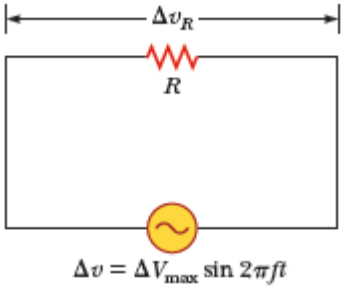
We conclude this chapter with a discussion of electromagnetic waves, which are composed of fluctuating electric and magnetic fields. Electromagnetic waves in the form of visible light enable us to view the world around us; infrared waves warm our environment; radiofrequency waves carry our television and radio programs, as well as information about processes in the core of our galaxy; and X-rays allow us to perceive structures hidden inside our bodies and study properties of distant, collapsed stars. Light is key to our understanding of the universe.

7.1 Resistors in an AC Circuit

An AC circuit consists of combinations of circuit elements and an AC generator or an AC source, which provides the alternating current. We have seen that the output of an AC generator is sinusoidal and varies with time according to

$$\Delta v = \Delta V_{\max} \sin 2\pi ft \qquad [7.1]$$

where Δv is the instantaneous voltage, ΔV_{max} is the maximum voltage of the AC generator, and f is the frequency at which the voltage changes, measured in hertz (Hz). We first consider a simple circuit consisting of a resistor and an AC source (designated by the symbol), as in Active Figure 7.1.



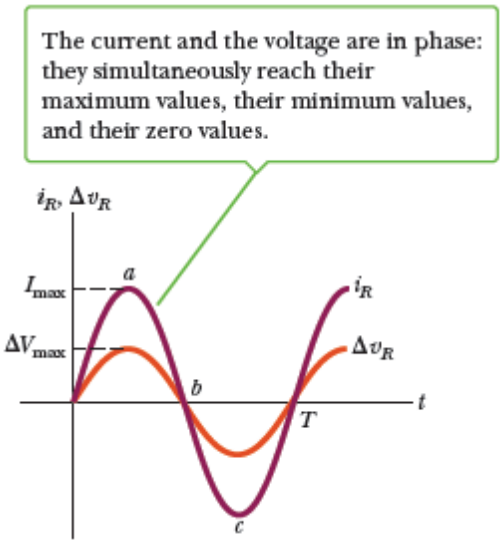
Active Figure 7.1

The current and the voltage versus time across the resistor are shown in Active Figure 7.2. To explain the concept of alternating current, we begin by discussing the current versus time curve in Active Figure 7.2. At point a on the curve, the current has a maximum value in one direction, arbitrarily called the positive direction. Between points a and b, the current is decreasing in magnitude but is still in the positive direction. At point b, the current is momentarily zero; it then begins to increase in the opposite (negative) direction between points b and c. At point c, the current has reached its maximum value in the negative direction. The current and voltage are in step with each other because they vary identically with time. **Because the current and the voltage reach their maximum values at the same time, they are said to be in phase**. Notice that **the average value of the current over one cycle is zero** because the current is maintained in one direction (the positive direction) for the same amount of time and at the same magnitude as it is in the opposite direction (the negative direction). The direction of the current, however, has no effect on the behavior of the resistor in the circuit: the collisions between electrons and the fixed atoms of the resistor result in an increase in the resistor's temperature regardless of the direction of the current. We can quantify this discussion by recalling that the rate at which electrical energy is dissipated in a resistor, the power P, is

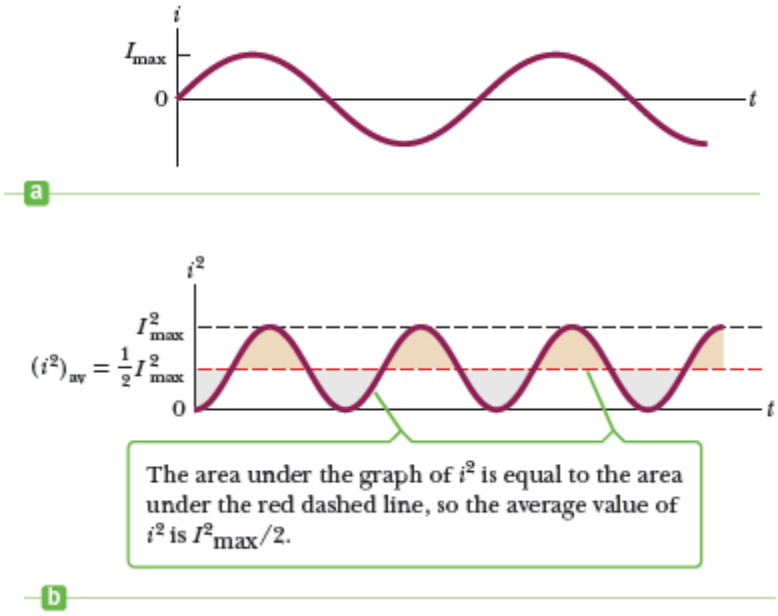
$$P = i^2 R$$

where i is the instantaneous current in the resistor. Because the heating effect of a current is proportional to the square of the current, it makes no difference whether the sign associated with the current is positive or negative. The heating effect produced by an alternating current with a maximum value of I_{max} is not the same as that produced by a direct current of the same value, however. The reason is that the alternating current has this maximum value for only an instant of time during a cycle. The important quantity in an AC circuit is a special kind of average value of current, called the *rms current*: the direct current that dissipates the same amount of energy in a resistor that is dissipated by

the actual alternating current. Finding the average of the alternating current i, depicted in Figure 7.3a, would not be useful because that average is zero, whereas the rms current is always positive. To find the rms current, we first square the current, then find its average value, and finally take the square root of this average value. Hence, the rms current is the square root of the average (mean) of the square of the current. Because i^2 varies as



Active Figure 7.2



Active Figure 7.3

$\sin^2 2\pi ft$, the average value of i^2 is $\frac{1}{2}I^2_{max}$ (Fig. 21.3b).1 Therefore, the rms current I_{rms} is related to the maximum value of the alternating current I_{max} by

$$I_{rms} = \frac{I_{max}}{\sqrt{2}} = 0.707 I_{max} \qquad [7.2]$$

This equation says that an alternating current with a maximum value of 3 A produces the same heating effect in a resistor as a direct current of $(\frac{3}{\sqrt{2}})$ A. We can therefore say that the average power dissipated in a resistor that carries alternating current I is

$$P_{av} = I_{rms}^2 R$$

Alternating voltages are also best discussed in terms of rms voltages, with a relationship identical to the preceding one,

$$\Delta V_{rms} = \frac{\Delta V_{max}}{\sqrt{2}} = 0.707\, \Delta V_{max} \qquad [7.3]$$

where ΔV_{rms} is the rms voltage and ΔV_{max} is the maximum value of the alternating voltage. When we speak of measuring an AC voltage of 120 V from an electric outlet, we actually mean an rms voltage of 120 V. A quick calculation using Equation 7.3 shows that such an AC voltage actually has a peak value of about 170 V. In this chapter we use rms values when discussing alternating currents and voltages. One reason is that AC ammeters and voltmeters are designed to read rms values. Further,
if we use rms values, many of the equations for alternating current will have the same form as those used in the study of direct-current (DC) circuits. Table 7.1 summarizes the notations used throughout this chapter.

	Voltage	Current
Instantaneous value	Δv	i
Maximum value	ΔV_{max}	I_{max}
rms value	ΔV_{rms}	I_{rms}

Table 7.1

Consider the series circuit in Figure 7.1 , consisting of a resistor connected to an AC generator. A resistor impedes the current in an AC circuit, just as it does in a DC circuit. Ohm's law is therefore valid for an AC circuit, and we have

$$\Delta V_{R,rms} = I_{rms} R \qquad [7.4a]$$

The rms voltage across a resistor is equal to the rms current in the circuit times the resistance. This equation is also true if maximum values of current and voltage are used:

$$\Delta V_{R,max} = I_{max} R \qquad [7.4b]$$

PROBLEM An AC voltage source has an output of $\Delta v = (2.00\times 10^2\ V) \sin 2\pi ft$. This source is connected to a 1.00×10^2 V resistor as in Figure 7.1. Find the rms voltage and rms current in the resistor.

STRATEGY Compare the expression for the voltage output just given with the general form, $\Delta v = \Delta V_{max} \sin 2\pi ft$, finding the maximum voltage. Substitute this result into the expression for the rms voltage.

SOLUTION

Obtain the maximum voltage by comparison of the given expression for the output with the general expression:

$$\Delta v = (2.00 \times 10^2\,\text{V}) \sin 2\pi ft \qquad \Delta v = \Delta V_{\max} \sin 2\pi ft$$
$$\rightarrow \quad \Delta V_{\max} = 2.00 \times 10^2\,\text{V}$$

Next, substitute into Equation 21.3 to find the rms voltage of the source:

$$\Delta V_{\text{rms}} = \frac{\Delta V_{\max}}{\sqrt{2}} = \frac{2.00 \times 10^2\,\text{V}}{\sqrt{2}} = 141\,\text{V}$$

Substitute this result into Ohm's law to find the rms current:

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{R} = \frac{141\,\text{V}}{1.00 \times 10^2\,\Omega} = 1.41\,\text{A}$$

7.2 Capacitors in an AC Circuit

To understand the effect of a capacitor on the behavior of a circuit containing an AC voltage source, we first review what happens when a capacitor is placed in a circuit containing a DC source, such as a battery. When the switch is closed in a series circuit containing a battery, a resistor, and a capacitor, the initial charge on the plates of the capacitor is zero. The motion of charge through the circuit is therefore relatively free, and there is a large current in the circuit. As more charge accumulates on the capacitor, the voltage across it increases, opposing the current. After some time interval, which depends on the time constant RC, the current approaches zero. Consequently, a capacitor in a DC circuit limits or impedes the current so that it approaches zero after a brief time.

Now consider the simple series circuit in Figure 7.4, consisting of a capacitor connected to an AC generator. We sketch curves of current versus time and voltage versus time, and then attempt to make the graphs seem reasonable. The curves are shown in Figure 7.5.

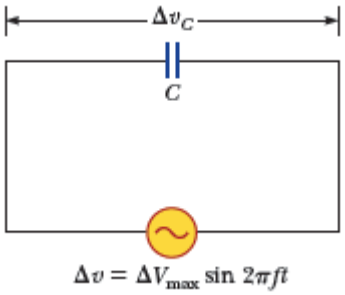


Figure 7.4

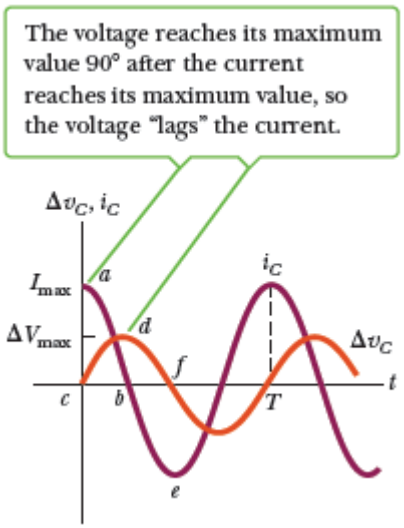


Figure 7.5

First, notice that the segment of the current curve from a to b indicates that the current starts out at a rather large value. This large value can be understood by recognizing that there is no charge on the capacitor at t = 0; as a consequence, there is nothing in the circuit except the resistance of the wires to hinder the flow of charge at this instant. The current decreases, however, as the voltage across the capacitor increases from c to d on the voltage curve. When the voltage is at point d, the current reverses and begins to increase in the opposite direction (from b to e on the current curve). During this time, the voltage across the capacitor decreases from d to f because the plates are now losing the charge they accumulated earlier. The remainder of the cycle for both voltage and current is a repeat of what happened during the first half of the cycle. The current reaches a maximum value in the opposite direction at point e on the current curve and then decreases as the voltage across the capacitor builds up. In a purely resistive circuit, the current and voltage are always in step with each

other. That isn't the case when a capacitor is in the circuit. In Figure 7.5, when an alternating voltage is applied across a capacitor, the voltage reaches its maximum value one-quarter of a cycle after the current reaches its maximum value. We say that the voltage across a capacitor always lags the current by 90°. The impeding effect of a capacitor on the current in an AC circuit is expressed in terms of a factor called the **capacitive reactance** X_C, defined as

$$X_C \equiv \frac{1}{2\pi f C} \qquad [7.5]$$

When C is in farads and f is in hertz, the unit of X_C is the ohm. Notice that $2\pi f = \omega$, the angular frequency. From Equation 7.5, as the frequency f of the voltage source increases, the capacitive reactance X_C (the impeding effect of the capacitor) decreases, so the current increases. At high frequency, there is less time available to charge the capacitor, so less charge and voltage accumulate on the capacitor, which translates into less opposition to the flow of charge and, consequently, a higher current. The analogy between capacitive reactance and resistance means that we can write an equation of the same form as Ohm's law to describe AC circuits containing capacitors. This equation relates the rms voltage and rms current in the circuit to the capacitive reactance:

$$\Delta V_{C,\text{rms}} = I_{\text{rms}} X_C \qquad [7.6]$$

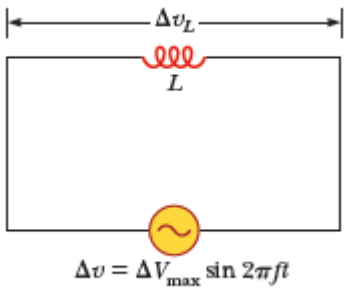
7.3 Inductors in an AC Circuit

Now consider an AC circuit consisting only of an inductor connected to the terminals of an AC source, as in Active Figure 7.6. (In any real circuit there is some resistance in the wire forming the inductive coil, but we ignore this consideration for now.) The changing current output of the generator produces a back emf that impedes the current in the circuit. The magnitude of this back emf is

$$\Delta v_L = L\frac{\Delta I}{\Delta t} \qquad [7.7]$$

The effective resistance of the coil in an AC circuit is measured by a quantity called the **inductive reactance**, X_L:

$$X_L \equiv 2\pi f L \qquad [7.8]$$



FFigure 7.6

When f is in hertz and L is in henries, the unit of X_L is the ohm. The inductive reactance increases with increasing frequency and increasing inductance. Contrast these facts with capacitors, where increasing frequency or capacitance decreases the capacitive reactance.

To understand the meaning of inductive reactance, compare Equation 7.8 with Equation 7.7. First, note from Equation 7.8 that the inductive reactance depends on the inductance L, which is reasonable because the back emf (Eq. 7.7) is large for large values of L. Second, note that the inductive reactance depends on the frequency f. This dependence, too, is reasonable because the back emf depends on $\Delta I/\Delta t$, a quantity that is large when the current changes rapidly, as it would for high frequencies. With inductive reactance defined in this way, we can write an equation of the same form as Ohm's law for the voltage across the coil or inductor:

$$\Delta V_{L,\mathrm{rms}} = I_{\mathrm{rms}} X_L \qquad [7.9]$$

where $\Delta V_{L,rms}$ is the rms voltage across the coil and I_{rms} is the rms current in the coil. Active Figure 7.7 shows the instantaneous voltage and instantaneous current across the coil as functions of time. When a sinusoidal voltage is applied across an inductor, the voltage reaches its maximum value one-quarter of an oscillation period before the current reaches its maximum value. In this situation we say that the voltage across an inductor always leads the current by 90°. To see why there is a phase relationship between voltage and current, we examine a few points on the curves of Active Figure 7.7. At point a on the current curve, the current is beginning to increase in the positive direction. At this instant the rate of change of current, $\Delta I/\Delta t$ (the slope of the current curve), is at a maximum, and we see from Equation 7.7 that the voltage across the inductor is consequently also at a maximum. As the current rises between points a and b on the curve, $\Delta I/\Delta t$ gradually decreases until it reaches zero at point b. As a result, the voltage across the inductor is decreasing during this same time interval, as the segment between c and d on the voltage curve indicates. Immediately after point b, the current begins to decrease, although it still has the same direction it had during the previous quarter cycle. As the current decreases to zero (from b to e on the curve), a voltage is again induced in the coil (from d to f), but the polarity of this voltage is opposite the polarity of the voltage induced between c and d. This occurs because back emfs always oppose the change in the current.

We could continue to examine other segments of the curves, but no new information would be gained because the current and voltage variations are repetitive.

7.4 The RLC Series Circuit

In the foregoing sections we examined the effects of an inductor, a capacitor, and a resistor when they are connected separately across an AC voltage source. We now consider what happens when these elements are combined. Active Figure 7.8 shows a circuit containing a resistor, an inductor, and a capacitor connected in series across an AC source that supplies a total voltage Δv at some instant.

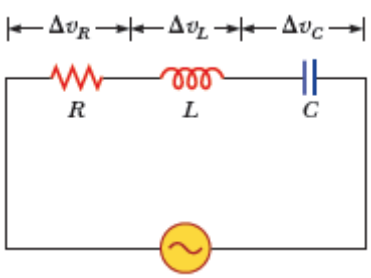


Figure 7.9

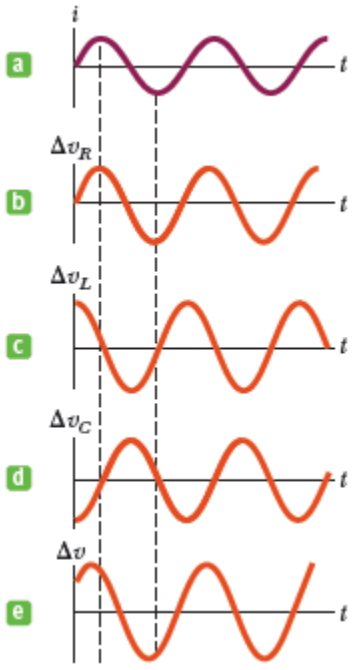


Figure 7.9

The current in the circuit is the same at all points in the circuit at any instant and varies sinusoidally with time, as indicated in Active Figure 7.9a. This fact can be expressed mathematically as

$$i = I_{\max} \sin 2\pi ft$$

Earlier, we learned that the voltage across each element may or may not be in phase with the current. The instantaneous voltages across the three elements, shown in Active Figure 7.9, have the following phase relations to the instantaneous current:

1. The instantaneous voltage Δv_R across the resistor is in phase with the instantaneous current. (See Active Fig. 7.9b.)
2. The instantaneous voltage Δv_L across the inductor leads the current by 90°. (See Active Fig. 7.9c.)
3. The instantaneous voltage Δv_C across the capacitor lags the current by 90°. (See Active Fig. 7.9d.

The net instantaneous voltage Δv supplied by the AC source equals the sum of the instantaneous voltages across the separate elements: $\Delta v = \Delta v_R + \Delta v_C + \Delta v_L$. This doesn't mean, however, that the voltages measured with an AC voltmeter across R, C, and L sum to the measured source voltage! In fact, the measured voltages don't sum to the measured source voltage because the voltages across R, C, and L all have different phases. To account for the different phases of the voltage drops, we use a technique involving vectors. We represent the voltage across each element with a rotating vector, as in Figure 7.10.

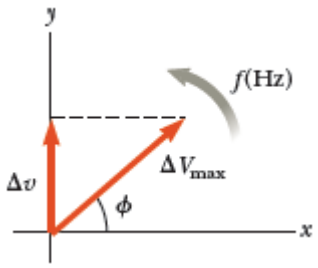


Figure 7.10

The rotating vectors are referred to as phasors, and the diagram is called a phasor diagram. This particular diagram represents the circuit voltage given by the expression $\Delta v = \Delta V_{max} \sin(2\pi ft + \phi)$, where ΔV_{max} is the maximum voltage (the magnitude or length of the rotating vector or phasor) and ϕ is the angle between the phasor and the positive x-axis when t = 0. The phasor can be viewed as a vector of magnitude ΔV_{max} rotating at a constant frequency f so that its projection along the y-axis is the instantaneous voltage in the circuit. Because f is the phase angle between the voltage and current in the circuit, the phasor for the current (not shown in Fig.7.10) lies along the positive x-axis when t = 0 and is expressed by the relation $i = I_{max} \sin(2\pi ft)$. The phasor diagrams in Figure 7.11 are useful for analyzing the series RLC circuit. Voltages in phase with the current are represented by vectors along the positive x-axis, and voltages out of phase with the current lie along other directions. ΔV_R is horizontal and to the right because it's in phase with the current. Likewise, ΔV_L is represented by a phasor along the positive y-axis because it leads the current by 90°. Finally, ΔV_C is along the negative y-axis because it lags the current by 90°. If the phasors are added as vector quantities so as to account for the different phases of the voltages across R, L, and C, Figure 7.11a shows that the only x-component for the voltages is ΔV_R and the net y-component is $\Delta V_L = \Delta V_C$. We now add the phasors vectorially to find the phasor ΔV_{max} (Fig. 7.11b), which represents the maximum voltage. The right triangle in Figure 7.11b gives the following equations for the maximum voltage and the phase angle f between the maximum voltage and the current:

$$\Delta V_{\max} = \sqrt{\Delta V_R{}^2 + (\Delta V_L - \Delta V_C)^2} \qquad [7.10]$$

$$\tan \phi = \frac{\Delta V_L - \Delta V_C}{\Delta V_R} \qquad [7.11]$$

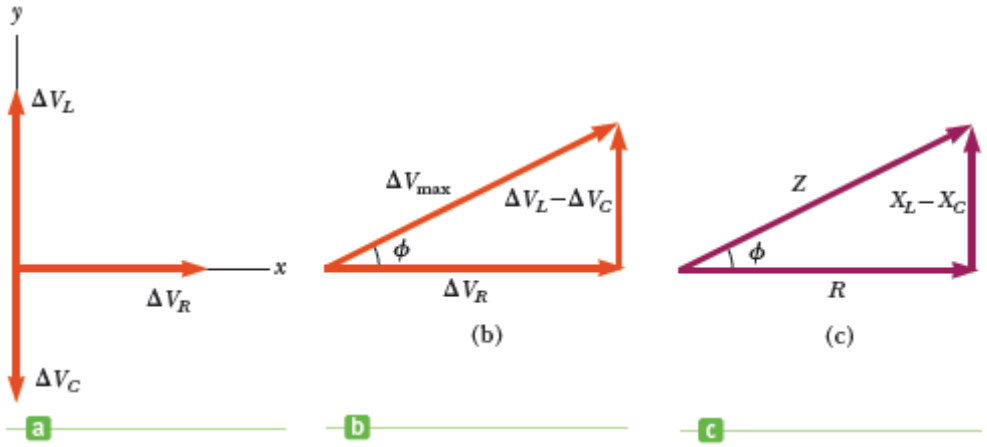


Figure 7.11

In these equations, all voltages are maximum values. Although we choose to use maximum voltages in our analysis, the preceding equations apply equally well to rms voltages because the two quantities are related to each other by the same factor for all circuit elements. The result for the maximum voltage ΔV_{max} given by Equation 7.10 reinforces the fact that **the voltages across the resistor, capacitor, and inductor are not in phase, so one cannot simply add them to get the voltage across the combination of element or to get the source voltage**.

We can write Equation 7.10 in the form of Ohm's law, using the relations $\Delta V_R = I_{max}R$, $\Delta V_L = I_{max}X_L$, and $\Delta V_C = I_{max}X_C$, where I_{max} is the maximum current in the circuit:

$$\Delta V_{\max} = I_{\max}\sqrt{R^2 + (X_L - X_C)^2} \qquad [7.12]$$

It's convenient to define a parameter called the **impedance** Z of the circuit as

$$Z \equiv \sqrt{R^2 + (X_L - X_C)^2} \qquad [7.13]$$

so that Equation 7.12 becomes

$$\Delta V_{\max} = I_{\max} Z \qquad [7.14]$$

Equation 7.14 is in the form of Ohm's law, $\Delta V = IR$, with R replaced by the impedance in ohms. Indeed, Equation 7.14 can be regarded as a generalized form of Ohm's law applied to a series AC circuit. Both the impedance and therefore the current in an AC circuit depend on the resistance, the inductance, the capacitance, and the frequency (because the reactances are frequency dependent).

It's useful to represent the impedance Z with a vector diagram such as the one depicted in Figure 7.11c. A right triangle is constructed with right side X_L - X_C, base R, and hypotenuse Z. Applying the Pythagorean theorem to this triangle, we see that

$$\tan\phi = \frac{X_L - X_C}{R} \qquad [7.15]$$

The physical significance of the phase angle will become apparent in Section 7.5. Table 7.2 provides impedance values and phase angles for some series circuits containing different combinations of circuit elements. Parallel alternating current circuits are also useful in everyday applications. We won't discuss them here, however, because their analysis is beyond the scope of this book.

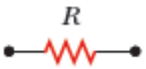
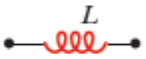
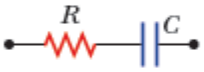
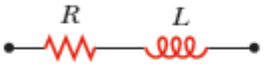
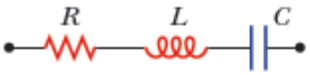
Circuit Elements	Impedance Z	Phase Angle ϕ
	R	0°
	X_C	−90°
	X_L	+90°
	$\sqrt{R^2 + X_C^2}$	Negative, between −90° and 0°
	$\sqrt{R^2 + X_L^2}$	Positive, between 0° and 90°
	$\sqrt{R^2 + (X_L - X_C)^2}$	Negative if $X_C > X_L$ Positive if $X_C < X_L$

Table 7.2

PROBLEM A series RLC AC circuit has resistance R = 2.50 × 10^2 Ω, inductance L = 0.600 H, capacitance C = 3.50 μF, frequency f = 60.0 Hz, and maximum voltage ΔV_{max} = 1.50 × 10^2 V. Find (a) the impedance of the circuit, (b) the maximum current in the circuit, (c) the phase angle, and (d) the maximum voltages across the elements.

STRATEGY Calculate the inductive and capacitive reactances, which can be used with the resistance to calculate the impedance and phase angle. The impedance and Ohm's law yield the maximum current.

SOLUTION

(a) Find the impedance of the circuit.

First, calculate the inductive and capacitive reactances:

$$X_L = 2\pi fL = 226\ \Omega \qquad X_C = 1/2\pi fC = 758\ \Omega$$

Substitute these results and the resistance R into Equation 21.13 to obtain the impedance of the circuit:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
$$= \sqrt{(2.50 \times 10^2\ \Omega)^2 + (226\ \Omega - 758\ \Omega)^2} = 588\ \Omega$$

(b) Find the maximum current in the circuit.

Use Equation 21.12, the equivalent of Ohm's law, to find the maximum current:

$$I_{max} = \frac{\Delta V_{max}}{Z} = \frac{1.50 \times 10^2\ \text{V}}{588\ \Omega} = 0.255\ \text{A}$$

(c) Find the phase angle.

Calculate the phase angle between the current and the voltage with Equation 21.15:

$$\phi = \tan^{-1}\frac{X_L - X_C}{R} = \tan^{-1}\left(\frac{226\ \Omega - 758\ \Omega}{2.50 \times 10^2\ \Omega}\right) = -64.8°$$

(d) Find the maximum voltages across the elements.

Substitute into the "Ohm's law" expressions for each individual type of current element:

$$\Delta V_{R,max} = I_{max}R = (0.255\ \text{A})(2.50 \times 10^2\ \Omega) = 63.8\ \text{V}$$
$$\Delta V_{L,max} = I_{max}X_L = (0.255\ \text{A})(2.26 \times 10^2\ \Omega) = 57.6\ \text{V}$$
$$\Delta V_{C,max} = I_{max}X_C = (0.255\ \text{A})(7.58 \times 10^2\ \Omega) = 193\ \text{V}$$

7.5 The RLC Series Circuit

No power losses are associated with pure capacitors and pure inductors in an AC circuit. A pure capacitor, by definition, has no resistance or inductance, whereas a pure inductor has no resistance or capacitance. (These definitions are idealizations: in a real capacitor, for example, inductive effects could become important at high frequencies.) We begin by analyzing the power dissipated in an AC circuit that contains only a generator and a capacitor. When the current increases in one direction in an AC circuit, charge accumulates on the capacitor and a voltage drop appears across it. When the voltage reaches its maximum value, the energy stored in the capacitor is

$$PE_C = \frac{1}{2}C(V_{max})^2$$

This energy storage is only momentary, however: When the current reverses direction, the charge leaves the capacitor plates and returns to the voltage source. During one-half of each cycle the capacitor is being charged, and during the other half the charge is being returned to the voltage source. Therefore, the average power supplied by the source is zero. In other words, no power losses occur in a capacitor in an AC circuit.

Similarly, the source must do work against the back emf of an inductor that is carrying a current. When the current reaches its maximum value, the energy stored in the inductor is a maximum and is given by

$$PE_L = \frac{1}{2}LI_{max}^2$$

When the current begins to decrease in the circuit, this stored energy is returned to the source as the inductor attempts to maintain the current in the circuit. The average power delivered to a resistor in an RLC circuit is

$$P_{av} = I_{max}^2 R \qquad [7.16]$$

The average power delivered by the generator is converted to internal energy in the resistor. No power loss occurs in an ideal capacitor or inductor. An alternate equation for the average power loss in an AC circuit can be found by substituting (from Ohm's law) $R = \Delta V_{R,rms}/I_{rms}$ into Equation 7.16:

$$P_{av} = I_{max}\Delta V_{R,rms}$$

It's convenient to refer to a voltage triangle that shows the relationship among ΔV_{rms}, $\Delta V_{R,rms}$, and $\Delta V_{L,rms}$ - $\Delta V_{C,rms}$, such as Figure 7.11b. (Remember that Fig.7.11 applies to both maximum and rms voltages.) From this figure, we see that the voltage drop across a resistor can be written in terms of the voltage of the source, ΔV_{rms}:

$$\Delta V_{R,rms} = \Delta V_{rms} \cos\phi$$

Hence, the average power delivered by a generator in an AC circuit is

$$P_{av} = I_{rms}\Delta V_{rms}\cos\phi \qquad [7.17]$$

where the quantity $\cos\phi$ is called the **power factor**.

Equation 7.17 shows that the power delivered by an AC source to any circuit depends on the phase difference between the source voltage and the resulting current. This fact has many interesting applications. For example, factories often use devices such as large motors in machines, generators,

and transformers that have a large inductive load due to all the windings. To deliver greater power to such devices without using excessively high voltages, factory technicians introduce capacitance in the circuits to shift the phase.

7.6 Resonance in a Series RLC Circuit

In general, the rms current in a series RLC circuit can be written

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{\Delta V_{\text{rms}}}{\sqrt{R^2 + (X_L - X_C)^2}} \qquad [7.18]$$

From this equation, we see that if the frequency is varied, the current has its maximum value when the impedance has its minimum value, which occurs when $X_L = X_C$. In such a circumstance, the impedance of the circuit reduces to $Z = R$. The frequency f_0 at which this happens is called the **resonance frequency** of the circuit. To find f_0, we set $X_L = X_C$, which gives, from Equations 7.5 and 7.8,

$$2\pi f_0 L = \frac{1}{2\pi f_0 C} \qquad f_0 = \frac{1}{2\pi\sqrt{LC}} \qquad [7.19]$$

Figure 7.13 is a plot of current as a function of frequency for a circuit containing a fixed value for both the capacitance and the inductance. From Equation 7.18, it must be concluded that the current would become infinite at resonance when $R = 0$. Although Equation 7.18 predicts this result, real circuits always have some resistance, which limits the value of the current.

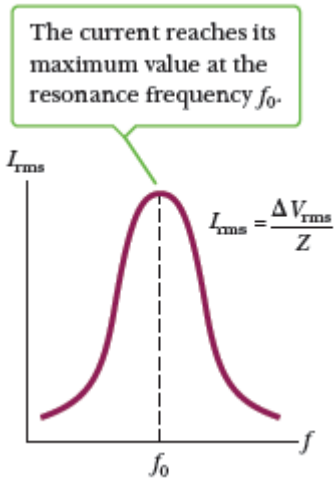


Figure 7.13

PROBLEM Consider a series RLC circuit for which $R = 1.50 \times 10^2\ \Omega$, $L = 20.0$ mH, $\Delta V_{\text{rms}} = 20.0$ V, and $f = 796\ \text{s}^{-1}$. (a) Determine the value of the capacitance for which the rms current is a maximum. (b) Find the maximum rms current in the circuit.

STRATEGY The current is a maximum at the resonance frequency f_0, which should be set equal to the driving frequency, $796\ \text{s}^{-1}$. The resulting equation can be solved for C. For part (b), substitute into Equation 7.18 to get the maximum rms current.

SOLUTION

(a) Find the capacitance giving the maximum current in the circuit (the resonance condition).

Solve the resonance frequency for the capacitance:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \rightarrow \sqrt{LC} = \frac{1}{2\pi f_0} \rightarrow LC = \frac{1}{4\pi^2 f_0{}^2}$$

$$C = \frac{1}{4\pi^2 f_0{}^2 L}$$

Insert the given values, substituting the source frequency for the resonance frequency, f_0:

$$C = \frac{1}{4\pi^2(796\ \text{Hz})^2(20.0 \times 10^{-3}\ \text{H})} = 2.00 \times 10^{-6}\ \text{F}$$

(b) Find the maximum rms current in the circuit.

The capacitive and inductive reactances are equal, so $Z = R = 1.50 \times 10^2\ \Omega$. Substitute into Equation 21.18 to find the rms current:

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{20.0\ \text{V}}{1.50 \times 10^2\ \Omega} = 0.133\ \text{A}$$

7.7 The Transformer and Power Transmission

It is economical to use a high voltage and a low current to minimize the I^2R loss in transmission lines when electric power is transmitted over great distances. Consequently, 350-kV lines are common, and in many areas, even higher-voltage (765-kV) lines are used. At the receiving end of such lines, the consumer requires power at a low voltage (for safety and for efficiency in design). In practice, the voltage is decreased to approximately 20 000 V at a distributing station, then to 4 000 V for delivery to residential areas, and finally to 120 V and 240 V at the customer's site. Therefore, a device is needed that can change the alternating voltage and current without causing appreciable changes in the power delivered. The AC transformer is that device.

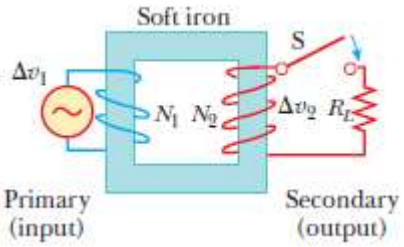


Figure 7.14

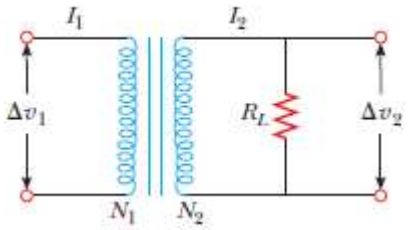


Figure 7.15

In its simplest form, the AC transformer consists of two coils of wire wound around a core of iron as illustrated in Figure 7.14. The coil on the left, which is connected to the input alternating voltage source and has N1 turns, is called the primary winding (or the primary). The coil on the right, consisting of N2 turns and connected to a load resistor RL , is called the secondary winding (or the secondary). The purposes of the iron core are to increase the magnetic flux through the coil and to provide a medium in which nearly all the magnetic field lines through one coil pass through the other coil. Eddy-current losses are reduced by using a laminated core. Transformation of energy to internal energy in the finite resistance of the coil wires is usually quite small. Typical transformers have power efficiencies from 90% to 99%. In the discussion that follows, let's assume we are working with an ideal transformer, one in which the energy losses in the windings and core are zero. Faraday's law states that the voltage Δv_1 across the primary is

$$\Delta v_1 = -N_1 \frac{d\Phi_B}{dt} \qquad [7.20]$$

where ϕ_B is the magnetic flux through each turn. If we assume all magnetic field lines remain within the iron core, the flux through each turn of the primary equals the flux through each turn of the secondary. Hence, the voltage across the secondary is

$$\Delta v_2 = -N_2 \frac{d\Phi_B}{dt} \qquad [7.21]$$

Solving Equation 7.20 for $d\phi_B/dt$ and substituting the result into Equation 7.21 gives

$$\Delta v_2 = \frac{N_2}{N_1} \Delta v_1 \qquad [7.22]$$

When N2 > N1, the output voltage Δv_2 exceeds the input voltage Δv1. This configuration is referred to as a step-up transformer. When N2 < N1, the output voltage is less than the input voltage, and we have a step-down transformer.

When the switch in the secondary circuit is closed, a current I_2 is induced in the secondary. (In this discussion, uppercase I and ΔV refer to rms values.) If the load in the secondary circuit is a pure resistance, the induced current is in phase with the induced voltage. The power supplied to the secondary circuit must be provided by the AC source connected to the primary circuit as shown in Figure 7.15. In an ideal transformer where there are no losses, the power $I_1 \Delta V_1$ supplied by the source is equal to the power $I_2 \Delta$V2 in the secondary circuit. That is,

$$I_1 \Delta V_1 = I_2 \Delta V_2 \qquad [7.23]$$

The value of the load resistance R_L determines the value of the secondary current because $I_2 \Delta V_2/R_L$. Furthermore, the current in the primary is $I_1 = \Delta V_1/R_{eq}$, where

$$R_{eq} = \left(\frac{N_1}{N_2}\right)^2 R_L \qquad [7.23]$$

is the equivalent resistance of the load resistance when viewed from the primary side. We see from this analysis that a transformer may be used to match resistances between the primary circuit and the load. In this manner, maximum power transfer can be achieved between a given power source and the load resistance. For example, a transformer connected between the 1-kΩ output of an audio amplifier and an 8Ω- speaker ensures that as much of the audio signal as possible is transferred into the speaker. In stereo terminology, this process is called *impedance matching*.

To operate properly, many common household electronic devices require low voltages. A small transformer that plugs directly into the wall like the one illustrated in Figure 7.16 can provide the proper voltage. The photograph shows the two windings wrapped around a common iron core that is found inside all these little “black boxes.” This particular transformer converts the 120-V AC in the wall socket to 12.5-V AC. (Can you determine the ratio of the numbers of turns in the two coils?) Some black boxes also make use of diodes to convert the alternating current to direct current.

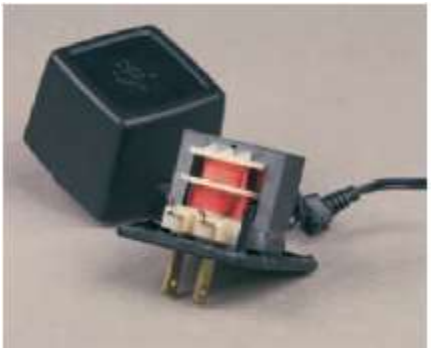


Figure 7.16

7.8 Maxwell's Predictions

During the early stages of their study and development, electric and magnetic phenomena were thought to be unrelated. In 1865, however, James Clerk Maxwell (1831–1879) provided a mathematical theory that showed a close relationship between all electric and magnetic phenomena. In addition to unifying the formerly separate fields of electricity and magnetism, his brilliant theory predicted that electric and magnetic fields can move through space as waves. The theory he developed is based on the following four pieces of information:

1. Electric field lines originate on positive charges and terminate on negative charges.
2. Magnetic field lines always form closed loops; they don't begin or end anywhere.
3. A varying magnetic field induces an emf and hence an electric field. This fact is a statement of Faraday's law.
4. Magnetic fields are generated by moving charges (or currents), as summarized in Ampère's law.

The first statement is a consequence of the nature of the electrostatic force between charged particles, given by Coulomb's law. It embodies the fact that **free charges (electric monopoles) exist in nature**.

The second statement—that magnetic fields form continuous loops—is exemplified by the magnetic field lines around a long, straight wire, which are closed circles, and the magnetic field lines of a bar magnet, which form closed loops. It says, in contrast to the first statement, that **free magnetic charges (magnetic monopoles) don't exist in nature**.

The third statement is equivalent to Faraday's law of induction, and the fourth is equivalent to Ampère's law.

In one of the greatest theoretical developments of the 19th century, Maxwell used these four statements within a corresponding mathematical framework to prove that electric and magnetic fields play symmetric roles in nature. It was already known from experiments that a changing magnetic field produced an electric field according to Faraday's law. Maxwell believed that nature was symmetric, and he therefore hypothesized that a changing electric field should produce
a magnetic field. This hypothesis could not be proven experimentally at the time it was developed because the magnetic fields generated by changing electric fields are generally very weak and therefore difficult to detect.

To justify his hypothesis, Maxwell searched for other phenomena that might be explained by it. He turned his attention to the motion of rapidly oscillating (accelerating) charges, such as those in a conducting rod connected to an alternating voltage. Such charges are accelerated and, according to Maxwell's predictions, generate changing electric and magnetic fields. The changing fields cause electromagnetic disturbances that travel through space as waves, similar to the spreading water waves created by a pebble thrown into a pool. The waves sent out by the oscillating charges are fluctuating electric and magnetic fields, so they are called electromagnetic waves. From Faraday's law and from Maxwell's own generalization of Ampère's law, Maxwell calculated the speed of the waves to be equal to the speed of light, $c = 3 \times 10^8$ m/s. He concluded that visible light and other electromagnetic waves consist of fluctuating electric and magnetic fields traveling through empty space, with each varying field inducing the other! His was truly one of the greatest discoveries of science, on a par with Newton's discovery of the laws of motion. Like Newton's laws, it had a profound influence on later scientific developments.

7.9 Hertz's Confirmation of Maxwell's Predictions

In 1887, after Maxwell's death, Heinrich Hertz (1857–1894) was the first to generate and detect electromagnetic waves in a laboratory setting, using LC circuits. In such a circuit a charged capacitor is connected to an inductor, as in Figure 7.17.

When the switch is closed, oscillations occur in the current in the circuit and in the charge on the capacitor. If the resistance of the circuit is neglected, no energy is dissipated and the oscillations continue.

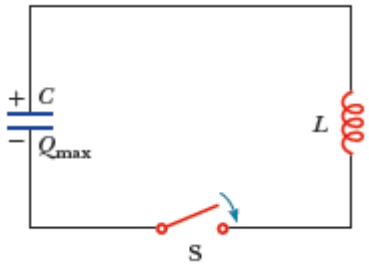


Figure 7.17

In the following analysis, we neglect the resistance in the circuit. We assume the capacitor has an initial charge of Q_{max} and the switch is closed at t = 0. When the capacitor is fully charged, the total energy in the circuit is stored in the electric field of the capacitor and is equal to $Q^2_{max}/2C$. At this time, the current is zero, so no energy is stored in the inductor. As the capacitor begins to discharge, the energy stored in its electric field decreases. At the same time, the current increases and energy equal to $LI^2/2$ is now stored in the magnetic field of the inductor. Thus, energy is transferred from the electric field of the capacitor to the magnetic field of the inductor. When the capacitor is fully discharged, it stores no energy. At this time, the current reaches its maximum value and all the energy is stored in the inductor. The process then repeats in the reverse direction. The energy continues to transfer between the inductor and the capacitor, corresponding to oscillations in the current and charge. As we saw in Section 7.6, the frequency of oscillation of an LC circuit is called the resonance frequency of the circuit and is given by

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

The circuit Hertz used in his investigations of electromagnetic waves is similar to that just discussed and is shown schematically in Figure 7.18. An induction coil (a large coil of wire) is connected to two metal spheres with a narrow gap between them to form a capacitor. Oscillations are initiated in the circuit by short voltage pulses sent via the coil to the spheres, charging one positive, the other negative. Because L and C are quite small in this circuit, the frequency of oscillation is quite high, f ≈ 100 MHz. This circuit is called a transmitter because it produces electromagnetic waves.

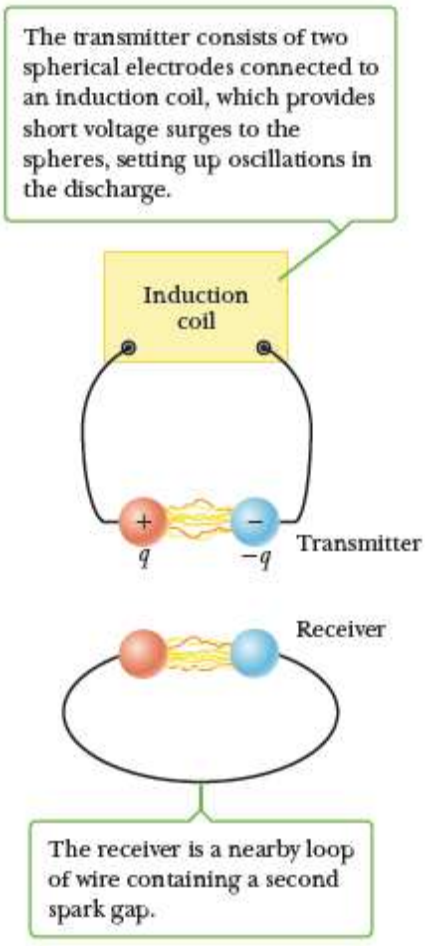


Figure 7.18

Several meters from the transmitter circuit, Hertz placed a second circuit, the receiver, which consisted of a single loop of wire connected to two spheres. It had its own effective inductance, capacitance, and natural frequency of oscillation. Hertz found that energy was being sent from the transmitter to the receiver when the resonance frequency of the receiver was adjusted to match that of the transmitter. The energy transfer was detected when the voltage across the spheres in the receiver circuit became high enough to produce ionization in the air, which caused sparks to appear in the air gap separating the spheres. Hertz's experiment is analogous to the mechanical phenomenon in which a tuning fork picks up the vibrations from another, identical tuning fork.

Hertz hypothesized that the energy transferred from the transmitter to the receiver is carried in the form of waves, now recognized as electromagnetic waves. In a series of experiments, he also showed that the radiation generated by the transmitter exhibits wave properties: interference, diffraction, reflection, refraction, and polarization. As you will see shortly, all these properties are exhibited by light. It became evident that Hertz's electromagnetic waves had the same known properties of light waves and differed only in frequency and wavelength. Hertz effectively confirmed Maxwell's theory by showing that Maxwell's mysterious electromagnetic waves existed and had all the properties of light waves.

Perhaps the most convincing experiment Hertz performed was the measurement of the speed of waves from the transmitter, accomplished as follows: waves of known frequency from the transmitter were reflected from a metal sheet so that an interference pattern was set up, much like the standing-wave pattern on a stretched string. As we learned in our discussion of standing waves, the distance between nodes is $\lambda/2$, so Hertz was able to determine the wavelength λ. Using the relationship $v = \lambda f$, he found that v was close to 3×10^8 m/s, the known speed of visible light. Hertz's experiments thus provided the first evidence in support of Maxwell's theory.

7.10 Production of Electromagnetic Waves by an Antenna

In the previous section we found that the energy stored in an LC circuit is continually transferred between the electric field of the capacitor and the magnetic field of the inductor. This energy transfer, however, continues for prolonged periods of time only when the changes occur slowly. If the current alternates rapidly, the circuit loses some of its energy in the form of electromagnetic waves. In fact, electromagnetic waves are radiated by any circuit carrying an alternating current. The fundamental mechanism responsible for this radiation is the acceleration of a charged particle. Whenever a charged particle accelerates, it radiates energy.

An alternating voltage applied to the wires of an antenna forces electric charges in the antenna to oscillate. This common technique for accelerating charged particles is the source of the radio waves emitted by the broadcast antenna of a radio station. Figure 7.19 illustrates the production of an electromagnetic wave by oscillating electric charges in an antenna. Two metal rods are connected to an AC source, which causes charges to oscillate between the rods. The output voltage of the generator is sinusoidal. At t = 0, the upper rod is given a maximum positive charge and the bottom rod an equal negative charge, as in Figure 7.19a. The electric field near the antenna at this instant is also shown in the figure. As the charges oscillate, the rods become less charged, the field near the rods decreases in strength, and the downward-directed maximum electric field produced at t = 0 moves away from the rod. When the charges are neutralized, as in Figure 21.18b, the electric field has dropped to zero, after an interval equal to one-quarter of the period of oscillation. Continuing in this fashion, the upper rod soon obtains a maximum negative charge and the lower rod becomes positive, as in Figure 7.19c, resulting in an electric field directed upward. This occurs after an interval equal to one-half the period of oscillation. The oscillations continue as indicated in Figure 7.19d

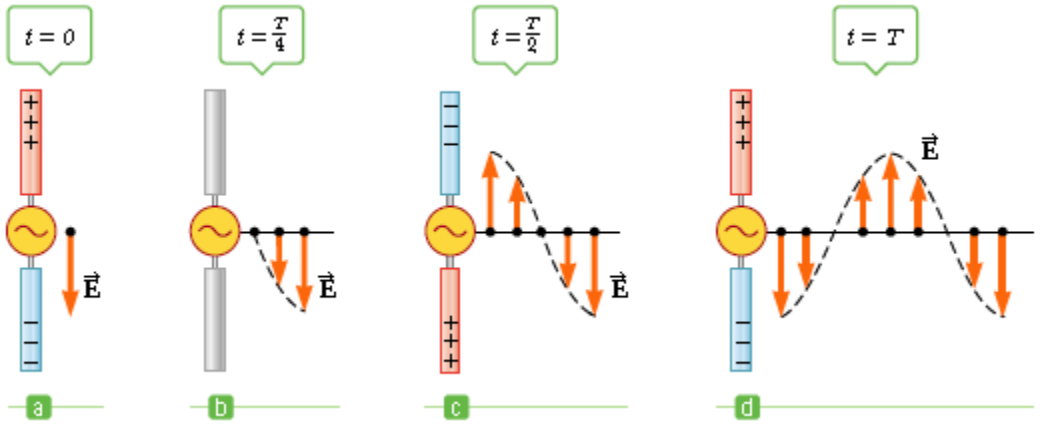


Figure 7.19

Note that the electric field near the antenna oscillates in phase with the charge distribution: the field points down when the upper rod is positive and up when the upper rod is negative. Further, the magnitude of the field at any instant depends on the amount of charge on the rods at that instant.

As the charges continue to oscillate (and accelerate) between the rods, the electric field set up by the charges moves away from the antenna in all directions at the speed of light. Figure 7.19 shows the electric field pattern on one side of the antenna at certain times during the oscillation cycle. As you can see, one cycle of charge oscillation produces one full wavelength in the electric field pattern.

Because the oscillating charges create a current in the rods, a magnetic field is also generated when the current in the rods is upward, as shown in Figure 7.20. The magnetic field lines circle the antenna (recall right-hand rule number 2) and are perpendicular to the electric field at all points. As the current changes with time, the magnetic field lines spread out from the antenna. At great distances from the antenna, the strengths of the electric and magnetic fields become very weak. At these distances, however, it is necessary to take into account the facts that (1) a changing magnetic field produces an electric field and (2) a changing electric field produces a magnetic field, as predicted by Maxwell. These induced electric and magnetic fields are in phase: at any point, the two fields reach their maximum values at the same instant. This synchrony is illustrated at one instant of time in Active Figure 7.20. Note that (1) the $\vec{E}$ and $\vec{B}$ fields are perpendicular to each other and (2) both fields are perpendicular to the direction of motion of the wave. This second property is characteristic of transverse waves. Hence, we see that an **electromagnetic wave is a transverse wave**.

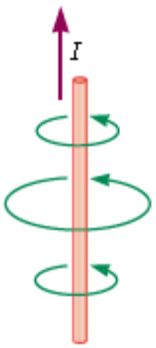


Figure 7.20

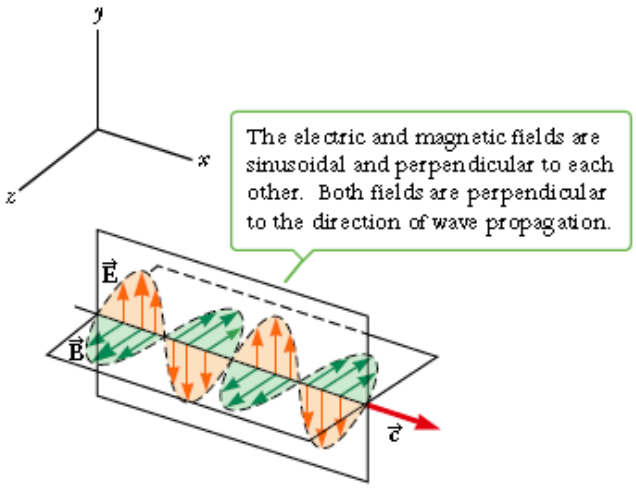


Figure 7.21

7.11 Properties of Electromagnetic Waves

We have seen that Maxwell's detailed analysis predicted the existence and properties of electromagnetic waves. In this section we summarize what we know about electromagnetic waves thus far and consider some additional properties. In our discussion here and in future sections, we will often make reference to a type of wave called a **plane wave**. A plane electromagnetic wave is a wave traveling from a very distant source. Active Figure 7.21 pictures such a wave at a given instant of time. In this case the oscillations of the electric and magnetic fields take place in planes perpendicular to the x-axis and are therefore perpendicular to the direction of travel of the wave. Because of the latter property, electromagnetic waves are transverse waves. In the figure the electric field $\vec{E}$ is in the y-direction and the magnetic field $\vec{B}$ is in the z-direction. Light propagates in a direction perpendicular to these two fields. That direction is determined by yet another right-hand rule: (1) point the fingers of your right hand in the direction of $\vec{E}$, (2) curl them in the direction of $\vec{B}$, and (3) the right thumb then points in the direction of propagation of the wave.

Electromagnetic waves travel with the speed of light. In fact, it can be shown that the speed of an electromagnetic wave is related to the permeability and permittivity of the medium through which it travels. Maxwell found this relationship for free space to be

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \qquad [7.24]$$

here c is the speed of light, μ_0 = 4π × 10-7 N.s2/C2 is the permeability constant of vacuum, and ε_0 = 8.85419 × 10-12 C2/N.m2 is the permittivity of free space.
Substituting these values into Equation 7.24, we find that

$$c = 2.997\,92 \times 10^8\ \mathrm{m/s} \qquad [7.25]$$

Because electromagnetic waves travel at the same speed as light in vacuum, scientists concluded (correctly) that **light is an electromagnetic wave**. Maxwell also proved the following relationship for electromagnetic waves:

$$\frac{E}{B} = c \qquad [7.26]$$

which states that the ratio of the magnitude of the electric field to the magnitude of the magnetic field equals the speed of light.

Electromagnetic waves carry energy as they travel through space, and this energy can be transferred to objects placed in their paths. The average rate at which energy passes through an area perpendicular to the direction of travel of a wave, or the average power per unit area, is called the **intensity** I of the wave and is given by

$$I = \frac{E_{\max} B_{\max}}{2\mu_0} \qquad [7.27]$$

where E_{max} and B_{max} are the maximum values of E and B. The quantity I is analogous to the intensity of sound waves. From Equation 7.26, we see that $E_{max} = cB_{max} = B_{max}/\sqrt{\mu_o \epsilon_o}$. Equation 7.27 can therefore also be expressed as

$$I = \frac{E_{max}^2}{2\mu_0 c} = \frac{c}{2\mu_0} B_{max}^2$$

[7.28]

Note that in these expressions we use the average power per unit area. A detailed analysis would show that the energy carried by an electromagnetic wave is shared equally by the electric and magnetic fields. Electromagnetic waves have an average intensity given by Equation 7.28. When the waves strike an area A of an object's surface for a given time Δt, energy $U = IA\Delta t$ is transferred to the surface. Momentum is transferred, as well. Hence, pressure is exerted on a surface when an electromagnetic wave impinges on it. In what follows, we assume the electromagnetic wave transports a total energy U to a surface in a time Δt. If the surface absorbs all the incident energy U in this time, Maxwell showed that the total momentum $\vec{p}$ delivered to this surface has a magnitude

$$p = \frac{U}{c} \qquad \text{(complete absorption)}$$

[7.29]

If the surface is a perfect reflector, then the momentum transferred in a time Δt for normal incidence is twice that given by Equation 7.29. This is analogous to a molecule of gas bouncing off the wall of a container in a perfectly elastic collision. If the molecule is initially traveling in the positive x-direction at velocity v and after the collision is traveling in the negative x-direction at velocity -v, its change in momentum is given by $\Delta p = mv - (-mv) = 2mv$. Light bouncing off a perfect reflector is a similar process, so for complete reflection,

$$p = \frac{2U}{c} \qquad \text{(complete reflection)}$$

[7.30]

Although radiation pressures are very small (about 5×10^{-6} N/m^2 for direct sunlight), they have been measured with a device such as the one shown in Figure 7.22.

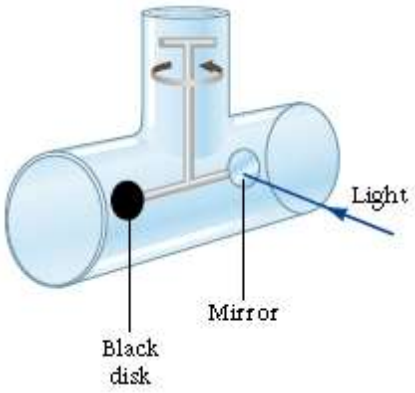


Figure 7.22

Light is allowed to strike a mirror and a black disk that are connected to each other by a horizontal bar suspended from a fine fiber. Light striking the black disk is completely absorbed, so all the momentum of the light is transferred to the disk. Light striking the mirror head-on is totally reflected; hence, the momentum transfer to the mirror is twice that transmitted to the disk. As a result, the horizontal bar supporting the disks twists counterclockwise as seen from above. The bar comes to equilibrium at some angle under the action of the torques caused by radiation pressure and the twisting of the fiber.

The radiation pressure can be determined by measuring the angle at which equilibrium occurs. The apparatus must be placed in a high vacuum to eliminate the effects of air currents. It's interesting that similar experiments demonstrate that electromagnetic waves carry angular momentum, as well.

In summary, electromagnetic waves traveling through free space have the following properties:

1. Electromagnetic waves travel at the speed of light.
2. Electromagnetic waves are transverse waves because the electric and magnetic fields are perpendicular to the direction of propagation of the wave and to each other.
3. The ratio of the electric field to the magnetic field in an electromagnetic wave equals the speed of light.
4. Electromagnetic waves carry both energy and momentum, which can be delivered to a surface.

7.12 The Spectrum of Electromagnetic Waves

All electromagnetic waves travel in a vacuum with the speed of light, c. These waves transport energy and momentum from some source to a receiver. In 1887 Hertz successfully generated and detected the radio-frequency electromagnetic waves predicted by Maxwell. Maxwell himself had recognized as electromagnetic waves both visible light and the infrared radiation discovered in 1800 by William Herschel. It is now known that other forms of electromagnetic waves exist that are distinguished by their frequencies and wavelengths.

Because all electromagnetic waves travel through free space with a speed c, their frequency f and wavelength l are related by the important expression

$$c = f\lambda \qquad [7.31]$$

The various types of electromagnetic waves are presented in Figure 7.23. Notice the wide and overlapping range of frequencies and wavelengths. For instance, an AM radio wave with a frequency of 1.50 MHz (a typical value) has a wavelength of

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{1.50 \times 10^6 \text{ s}^{-1}} = 2.00 \times 10^2 \text{ m}$$

The following abbreviations are often used to designate short wavelengths and distances:

1 micrometer (μm) = 10^{-6} m
1 nanometer (nm) = 10^{-9} m
1 angstrom (Å) = 10^{-10} m

The wavelengths of visible light, for example, range from 0.4 mm to 0.7 mm, or 400 nm to 700 nm, or 4 000 Å to 7 000 Å.

Quick Quiz

Which of the following statements are true about light waves? (a) The higher the frequency, the longer the wavelength. (b) The lower the frequency, the longer the wavelength. (c) Higher-frequency light travels faster than lower-frequency light. (d) The shorter the wavelength, the higher the frequency. (e) The lower the frequency, the shorter the wavelength.

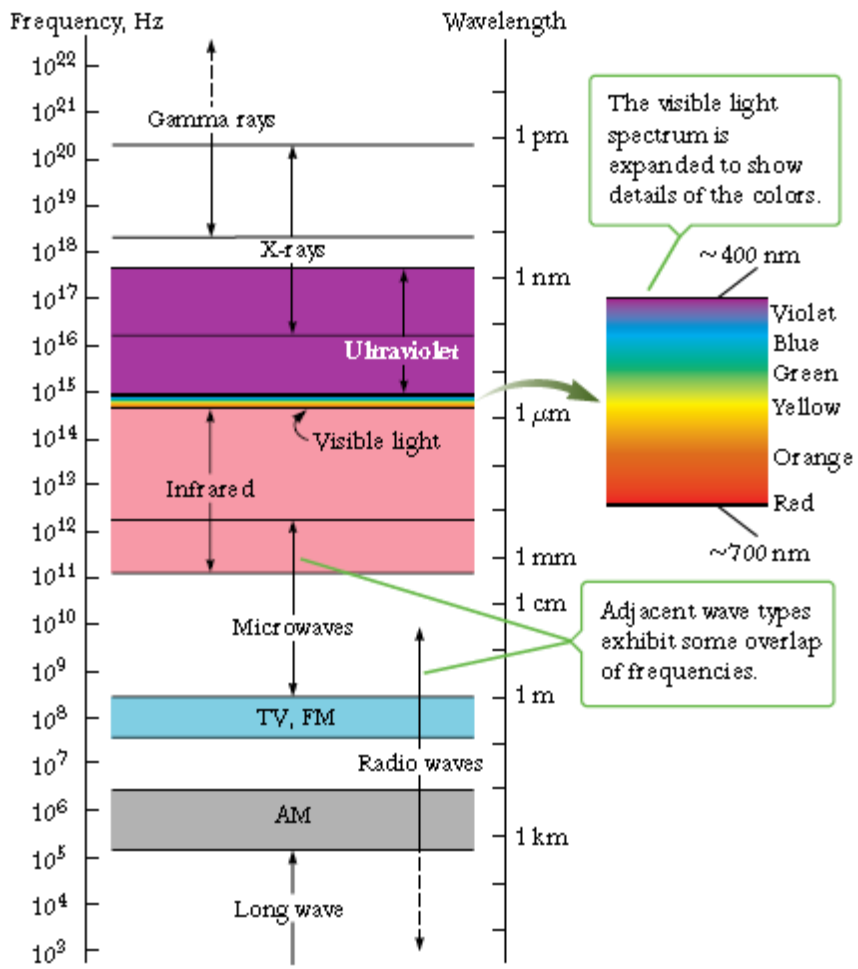


Figure 7.23

Brief descriptions of the wave types follow, in order of decreasing wavelength. There is no sharp division between one kind of electromagnetic wave and the next. All forms of electromagnetic radiation are produced by accelerating charges.

Radio waves, which were discussed in Section 7.10, are the result of charges accelerating through conducting wires. They are, of course, used in radio and television communication systems.

Microwaves (short-wavelength radio waves) have wavelengths ranging between about 1 mm and 30 cm and are generated by electronic devices. Their short wavelengths make them well suited for the radar systems used in aircraft navigation and for the study of atomic and molecular properties of matter. Microwave ovens are an interesting domestic application of these waves. It has been suggested that solar energy might be harnessed by beaming microwaves to Earth from a solar collector in space.

Infrared waves (sometimes incorrectly called "heat waves"), produced by hot objects and molecules, have wavelengths ranging from about 1 mm to the longest wavelength of visible light, 7×10^{-7} m. They are readily absorbed by most materials. The infrared energy absorbed by a substance causes it to get warmer because the energy agitates the atoms of the object, increasing their vibrational or translational motion. The result is a rise in temperature. Infrared radiation has many practical and scientific applications, including physical therapy, infrared photography, and the study of the vibrations of atoms.

Visible light, the most familiar form of electromagnetic waves, may be defined as the part of the spectrum that is detected by the human eye. Light is produced by the rearrangement of electrons in atoms and molecules. The wavelengths of visible light are classified as colors ranging from violet ($\lambda \approx 4 \times 10^{-7}$ m) to red ($\lambda \approx 7 \times 10^{-7}$ m). The eye's sensitivity is a function of wavelength and is greatest at a wavelength of about 5.6×10^{-7} m (yellow green).

Ultraviolet (UV) light covers wavelengths ranging from about 4×10^{-7} m (400 nm) down to 6×10^{-10} m (0.6 nm). The Sun is an important source of ultraviolet light (which is the main cause of suntans). Most of the ultraviolet light from the Sun is absorbed by atoms in the upper atmosphere, or stratosphere, which is fortunate, because UV light in large quantities has harmful effects on humans. One important constituent of the stratosphere is ozone (O_3), produced from reactions of oxygen with ultraviolet radiation. The resulting ozone shield causes lethal high-energy ultraviolet radiation to warm the stratosphere.

X-rays are electromagnetic waves with wavelengths from about 10^{-8} m (10 nm) down to 10^{-13} m (10^{-4} nm). The most common source of x-rays is the acceleration of high-energy electrons bombarding a metal target. X-rays are used as a diagnostic tool in medicine and as a treatment for certain forms of cancer. Because x-rays easily penetrate and damage or destroy living tissues and organisms, care must be taken to avoid unnecessary exposure and overexposure.

Gamma rays—electromagnetic waves emitted by radioactive nuclei—have wavelengths ranging from about 10^{-10} m to less than 10^{-14} m. They are highly penetrating and cause serious damage when absorbed by living tissues. Accordingly, those working near such radiation must be protected by garments containing heavily absorbing materials, such as layers of lead. When astronomers observe the same celestial object using detectors sensitive to different regions of the electromagnetic spectrum, striking variations in the

Light and Wound Treatment

An important issue in human health is wound management. Chronic wounds affect five million to seven million people in the United States at an annual cost of more than twenty billion dollars. Low-level laser therapy has been shown to facilitate the healing and closure of wounds. Infrared light increases the generation of adenosine triphosphate (ATP) in mitochondria and may stimulate the activation of genes and enzymes associated with cellular respiration. (ATP molecules provide energy for a variety of important cell functions.) Infrared light may also increase the concentration of reactive oxygen molecules, which could increase communication between the nucleus, cytosol, and the mitochondria. This mechanism may enhance and accelerate the healing process. Green laser light can also be used to stimulate the body's repair mechanisms via a different mechanism. A pink dye, called "rose bengal," is applied to the tissue, which is then exposed to the laser light for a few minutes. When the dye absorbs the light, it causes cross-linkages between collagen molecules in the tissue. The cross-linked molecules promote the closing of the tissue while reducing or eliminating the formation of scar tissue.



Figure 7.25

Figure 7.25 shows the contrast between tissue receiving the normal treatment and tissue irradiated by lasers. The technique is also being studied for application to damaged peripheral nerves, blood vessels, and other tissues, such as incisions made in the cornea during eye surgery.

The Sun and the Evolution of the Eye

The center of sensitivity of our eyes coincides with the center of the wavelength distribution of the Sun. Is this an amazing coincidence?

EXPLANATION This fact is not a coincidence; rather, it's the result of biological evolution. Humans have evolved with vision most sensitive to wavelengths that are strongest from the Sun. If aliens from another planet ever arrived at Earth, their eyes would have the center of sensitivity at wavelengths different from ours. If their sun were a red dwarf, for example, the alien's eyes would be most sensitive to red light.

PROBLEMS

1. (a) What is the resistance of a lightbulb that uses an average power of 75.0 W when connected to a 60.0-Hz power source having a maximum voltage of 170 V? (b) What is the resistance of a 100-W lightbulb?
2. A certain lightbulb is rated at 60.0 W when operating at an rms voltage of 120 V. (a) What is the peak voltage applied across the bulb? (b) What is the resistance of the bulb? (c) Does a 100-W bulb have greater or less resistance than a 60.0-W bulb? Explain.
3. The current in the circuit shown in Figure P7.1 equals 60.0% of the peak current at t 5 7.00 ms. What is the lowest source frequency that gives this current?

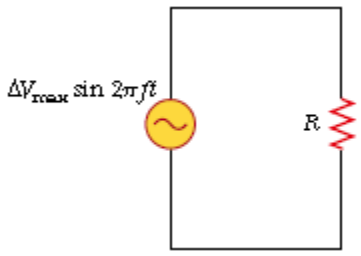


Figure P7.1

4. Figure P7.2 shows three lamps connected to a 120-V AC (rms) household supply voltage. Lamps 1 and 2 have 150-W bulbs; lamp 3 has a 100-W bulb. For each bulb, find (a) the rms current and (b) the resistance

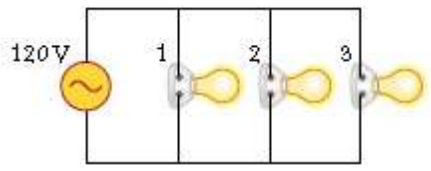


Figure P7.2

5. An audio amplifier, represented by the AC source and the resistor R in Figure P7.3, delivers alternating voltages at audio frequencies to the speaker. If the source puts out an

alternating voltage of 15.0 V (rms), the resistance R is 8.20 V, and the speaker is equivalent to a resistance of 10.4 V, what is the time-averaged power delivered to the speaker?

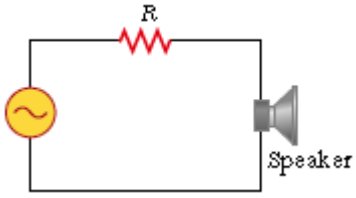


Figure P7.3

6. The output voltage of an AC generator is given by $\Delta v = (170\ V) \sin(60\pi t)$. The generator is connected across a 20.0-V resistor. By inspection, what are the (a) maximum voltage and (b) frequency? Find the (c) rms voltage across the resistor, (d) rms current in the resistor, (e) maximum current in the resistor, (f) power delivered to the resistor, and (g) current when t = 0.005 0 s. (h) Should the argument of the sine function be in degrees or radians?

7. (a) For what frequencies does a 22.0-μF capacitor have a reactance below 175 V? (b) What is the reactance of a 44.0-μF capacitor over this same frequency range?
8. What is the maximum current delivered to a circuit containing a 2.20-μF capacitor when it is connected across (a) a North American outlet having ΔV_{rms} =120 V and f = 60.0 Hz and (b) a European outlet having ΔV_{rms} = 240 V and f = 50.0 Hz?
9. When a 4.0-μF capacitor is connected to a generator whose rms output is 30 V, the current in the circuit is observed to be 0.30 A. What is the frequency of the source?
10. An AC generator with an output rms voltage of 36.0 V at a frequency of 60.0 Hz is connected across a 12.0-μF capacitor. Find the (a) capacitive reactance, (b) rms current, and (c) maximum current in the circuit. (d) Does the capacitor have its maximum charge when the current takes its maximum value? Explain.
11. What maximum current is delivered by an AC source with ΔV_{max} = 48.0 V and f = 90.0 Hz when connected across a 3.70-μF capacitor?
12. A generator delivers an AC voltage of the form $\Delta v = (98.0\ V) \sin(80\pi t)$ to a capacitor. The maximum current in the circuit is 0.500 A. Find the (a) rms voltage of the generator, (b) frequency of the generator, (c) rms current, (d) reactance, and (e) value of the capacitance.
13. An inductor has a 54.0-V reactance when connected to a 60.0-Hz source. The inductor is removed and then connected to a 50.0-Hz source that produces a 100-V rms voltage. What is the maximum current in the inductor?
14. An AC generator has an output rms voltage of 78.0 V at a frequency of 80.0 Hz. If the generator is connected across a 25.0-mH inductor, find the (a) inductive reactance, (b) rms current, and (c) maximum current in the circuit.
15. In a purely inductive AC circuit as shown in Figure P7.4, ΔV_{max} = 100 V. (a) The maximum current is 7.50 A at 50.0 Hz. Calculate the inductance L. (b) At what angular frequency ω is the maximum current 2.50 A?

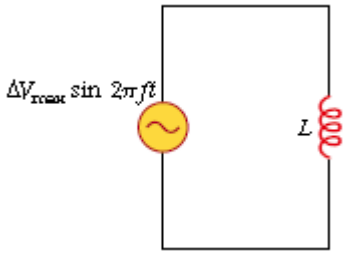


Figure P7.5

16. The output voltage of an AC generator is given by $\Delta v = (1.20 \times 10^2\ V) \sin(30\pi t)$. The generator is connected across a 0.500-H inductor. Find the (a) frequency of the generator,

(b) rms voltage across the inductor, (c) inductive reactance, (d) rms current in the inductor, (e) maximum current in the inductor, and (f) average power delivered to the inductor. (g) Find an expression for the instantaneous current. (h) At what time after t = 0 does the instantaneous current first reach 1.00 A? (Use the inverse sine function.)

17. Determine the maximum magnetic flux through an inductor connected to a standard outlet (ΔV_{rms} = 120 V, f = 60.0 Hz).
18. A sinusoidal voltage Δv = (80.0 V)sin(150t) is applied to a series RLC circuit with L = 80.0 mH, C = 125.0 μF, and R = 40.0 Ω. (a) What is the impedance of the circuit? (b) What is the maximum current in the circuit?
19. A 40.0-mF capacitor is connected to a 50.0-Ω resistor and a generator whose rms output is 30.0 V at 60.0 Hz. Find (a) the rms current in the circuit, (b) the rms voltage drop across the resistor, (c) the rms voltage drop across the capacitor, and (d) the phase angle for the circuit.
20. An inductor (L = 400 mH), a capacitor (C = 4.43 μF), and a resistor (R = 500 Ω) are connected in series. A 50.0-Hz AC generator connected in series to these elements produces a maximum current of 250 mA in the circuit. (a) Calculate the required maximum voltage ΔV_{max}. (b) Determine the phase angle by which the current leads or lags the applied voltage.
21. A resistor (R = 9.00 × 10^2 Ω), a capacitor (C = 0.250 μF), and an inductor (L = 2.50 H) are connected in series across a 2.40 × 10^2-Hz AC source for which ΔV_{max} = 1.40 × 10^2 V. Calculate (a) the impedance of the circuit, (b) the maximum current delivered by the source, and (c) the phase angle between the current and voltage. (d) Is the current leading or lagging the voltage?
22. A 50.0-Ω resistor, a 0.100-H inductor, and a 10.0-μF capacitor are connected in series to a 60.0-Hz source. The rms current in the circuit is 2.75 A. Find the rms voltages across (a) the resistor, (b) the inductor, (c) the capacitor, and (d) the RLC combination. (e) Sketch the phasor diagram for this circuit.
23. An RLC circuit consists of a 150-Ω resistor, a 21.0-μF capacitor, and a 460-mH inductor connected in series with a 120-V, 60.0-Hz power supply. (a) What is the phase angle between the current and the applied voltage? (b) Which reaches its maximum earlier, the current or the voltage?
24. An AC source operating at 60 Hz with a maximum voltage of 170 V is connected in series with a resistor (R = 1.2 kΩ) and an inductor (L = 2.8 H). (a) What is the maximum value of the current in the circuit? (b) What are the maximum values of the potential difference across the resistor and the inductor? (c) When the current is at a maximum, what are the magnitudes of the potential differences across the resistor, the inductor, and the AC source? (d) When the current is zero, what are the magnitudes of the potential difference across the resistor, the inductor, and the AC source?
25. A person is working near the secondary of a transformer, as shown in Figure P7.6. The primary voltage is 120 V (rms) at 60.0 Hz. The capacitance C_s, which is the stray capacitance between the hand and the secondary winding, is 20.0 pF. Assuming the person has a body resistance to ground of R_b = 50.0 kV, determine the rms voltage across the body. Hint: Redraw the circuit with the secondary of the transformer as a simple AC source.

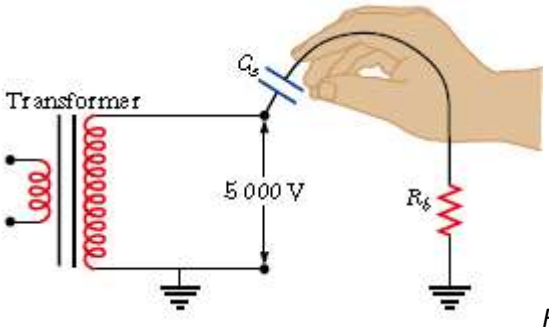


Figure P7.6

26. A 60.0-V resistor is connected in series with a 30.0-mF capacitor and a generator having a maximum voltage of 1.20×10^2 V and operating at 60.0 Hz. Find the (a) capacitive reactance of the circuit, (b) impedance of the circuit, and (c) maximum current in the circuit. (d) Does the voltage lead or lag the current? (e) How will putting an inductor in series with the existing capacitor and resistor affect the current? Explain
27. A series AC circuit contains a resistor, an inductor of 150 mH, a capacitor of 5.00 μF, and a generator with ΔV_{max} = 240 V operating at 50.0 Hz. The maximum current in the circuit is 100 mA. Calculate (a) the inductive reactance, (b) the capacitive reactance, (c) the impedance, (d) the resistance in the circuit, and (e) the phase angle between the current and the generator voltage.
28. At what frequency does the inductive reactance of a 57.0-mH inductor equal the capacitive reactance of a 57.0-μF capacitor?
29. An AC source with a maximum voltage of 150 V and f = 50.0 Hz is connected between points a and d in Figure P7.7. Calculate the rms voltages between points (a) a and b, (b) b and c, (c) c and d, and (d) b and d.

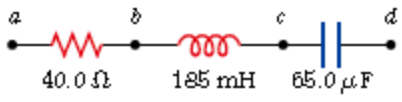


Figure P7.7

30. An AC source operating at 60 Hz with a maximum voltage of 170 V is connected in series with a resistor (R = 1.2 kΩ) and a capacitor (C = 2.5 μF). (a) What is the maximum value of the current in the circuit? (b) What are the maximum values of the potential difference across the resistor and the capacitor? (c) When the current is zero, what are the magnitudes of the potential difference across the resistor, the capacitor, and the AC source? How much charge is on the capacitor at this instant? (d) When the current is at a maximum, what are the magnitudes of the potential differences across the resistor, the capacitor, and the AC source? How much charge is on the capacitor at this instant?
31. A multimeter in an RL circuit records an rms current of 0.500 A and a 60.0-Hz rms generator voltage of 104 V. A wattmeter shows that the average power delivered to the resistor is 10.0 W. Determine (a) the impedance in the circuit, (b) the resistance R, and (c) the inductance L.
32. An AC voltage of the form Δv = (90.0 V) sin (350t) is applied to a series RLC circuit. If R = 50.0 Ω, C = 25.0 μF, and L = 0.200 H, find the (a) impedance of the circuit, (b) rms current in the circuit, and (c) average power delivered to the circuit.
33. An AC voltage of the form Δv = 100 sin (1 000t), where Δv is in volts and t is in seconds, is applied to a series RLC circuit. Assume the resistance is 400 V, the capacitance is 5.00 μF, and the inductance is 0.500 H. Find the average power delivered to the circuit.
34. A series RLC circuit has a resistance of 22.0 Ω and an impedance of 80.0 V. If the rms voltage applied to the circuit is 160 V, what average power is delivered to the circuit?

35. An inductor and a resistor are connected in series. When connected to a 60-Hz, 90-V (rms) source, the voltage drop across the resistor is found to be 50 V (rms) and the power delivered to the circuit is 14 W. Find (a) the value of the resistance and (b) the value of the inductance.
36. Consider a series RLC circuit with R = 25 Ω, L = 6.0 mH, and C = 25 μF. The circuit is connected to a 10-V (rms), 600-Hz AC source. (a) Is the sum of the voltage drops across R, L, and C equal to 10 V (rms)? (b) Which is greatest, the power delivered to the resistor, to the capacitor, or to the inductor? (c) Find the average power delivered to the circuit.
37. An RLC circuit is used in a radio to tune into an FM station broadcasting at f = 99.7 MHz. The resistance in the circuit is R = 12.0 V, and the inductance is L = 1.40 mH. What capacitance should be used?
38. The resonant frequency of a certain series RLC circuit is 2.84 kHz, and the value of its capacitance is 6.50 μF. What is the value of the resonant frequency when the capacitance of the circuit is 9.80 μF?
39. The AM band extends from approximately 500 kHz to 1 600 kHz. If a 2.0-mH inductor is used in a tuning circuit for a radio, what are the extremes that a capacitor must reach to cover the complete band of frequencies?
40. Consider a series RLC circuit with R = 15 V, L = 200 mH, C = 75 mF, and a maximum voltage of 150 V. (a) What is the impedance of the circuit at resonance? (b) What is the resonance frequency of the circuit? (c) When will the current be greatest: at resonance, at 10% below the resonant frequency, or at 10% above the resonant frequency? (d) What is the rms current in the circuit at a frequency of 60 Hz?
41. Two electrical oscillators are used in a heterodyne metal detector to detect buried metal objects (see Fig. P7.8). The detector uses two identical electrical oscillators in the form of LC circuits having resonant frequencies of 725 kHz. When the signals from the two oscillating circuits are combined, the beat frequency is zero because each has the same resonant frequency. However, when the coil of one circuit encounters a buried metal object, the inductance of this circuit increases by 1.000%, while that of the second is unchanged. Determine the beat frequency that would be detected in this situation.



Figure P7.8

42. A series circuit contains a 3.00-H inductor, a 3.00-μF capacitor, and a 30.0-Ω resistor connected to a 120-V (rms) source of variable frequency. Find the power delivered to the circuit when the frequency of the source is (a) the resonance frequency, (b) one-half the resonance frequency, (c) one-fourth the resonance frequency, (d) two times the resonance frequency, and (e) four times the resonance frequency. From your calculations,

can you draw a conclusion about the frequency at which the maximum power is delivered to the circuit?

43. The primary coil of a transformer has N_1 =250 turns, and its secondary coil has N_2 = 1 500 turns. If the input voltage across the primary coil is Δv = (170 V) sin ωt, what rms voltage is developed across the secondary coil?
44. A step-down transformer is used for recharging the batteries of portable devices. The turns ratio N_2/N_1 for a particular transformer used in a CD player is 1:13. When used with 120-V (rms) household service, the transformer draws an rms current of 250 mA. Find the (a) rms output voltage of the transformer and (b) power delivered to the CD player.
45. An AC power generator produces 50 A (rms) at 3 600 V. The voltage is stepped up to 100000 V by an ideal transformer, and the energy is transmitted through a long-distance power line that has a resistance of 100 V. What percentage of the power delivered by the generator is dissipated as heat in the power line?
46. A transformer is to be used to provide power for a computer disk drive that needs 6.0 V (rms) instead of the 120 V (rms) from the wall outlet. The number of turns in the primary is 400, and it delivers 500 mA (the secondary current) at an output voltage of 6.0 V (rms). (a) Should the transformer have more turns in the secondary compared with the primary, or fewer turns? (b) Find the current in the primary. (c) Find the number of turns in the secondary.
47. A transformer on a pole near a factory steps the voltage down from 3 600 V (rms) to 120 V (rms). The transformer is to deliver 1 000 kW to the factory at 90% efficiency. Find (a) the power delivered to the primary, (b) the current in the primary, and (c) the current in the secondary.
48. A transmission line that has a resistance per unit length of 4.50 3 1024 V/m is to be used to transmit 5.00 MW over 400 miles (6.44×10^5 m). The output voltage of the generator is 4.50 kV (rms). (a) What is the line loss if a transformer is used to step up the voltage to 500 kV (rms)? (b) What fraction of the input power is lost to the line under these circumstances? (c) What difficulties would be encountered on attempting to transmit the 5.00 MW at the generator voltage of 4.50 kV (rms)?

